

Group Office Hours

Feb. 20, 2021

ECON 301

Spring 2021



Task 1

$$(a) \quad E[\bar{Y} | X's] = E\left[\frac{1}{n} \sum_i Y_i | X's\right]$$

Linearity of
Expectation
Thm. 2.1

Strong
Exogeneity

A.2.

$$= \frac{1}{n} E\left[\sum_i Y_i | X's\right]$$

$$= \frac{1}{n} \sum_i E[Y_i | X's]$$

$$= \frac{1}{n} \sum_i E[Y_i | X_i]$$

$$= \frac{1}{n} \sum_i (\beta_0 + \beta_1 X_i)$$

$$= \beta_0 + \beta_1 \left(\frac{1}{n} \sum X_i\right)$$

$$= \beta_0 + \beta_1 \bar{X}$$

Task 6:

Be careful! From example 2.1 (Lecture 2)

$$\text{for } f(x, y) = (x + y) \mathbb{1}_{[0,1]}(x) \mathbb{1}_{[0,1]}(y)$$

$$f(y|x) = \frac{(x+y)}{(x+1/2)} \mathbb{1}_{[0,1]}(y)$$

$$\mathbb{E}[Y|X=x] = \frac{3x+2}{6x+3} \quad ! \text{ This is } \underline{\text{not}} \text{ linear}$$

$$\begin{aligned} \hat{\beta}_1 &= \frac{s_{xy}}{s_{xx}} = \frac{1}{s_{xx}} \sum (x_i - \bar{x})(y_i - \bar{y}) \\ &= \frac{1}{s_{xx}} \sum (x_i - \bar{x}) y_i \end{aligned}$$

$$E[\hat{\beta}_1 | X's] = E\left[\frac{s_{xy}}{s_{xx}} | X's\right]$$

$$= \frac{1}{s_{xx}} E\left[\sum (x_i - \bar{x}) y_i | X's\right]$$

$$= \frac{1}{s_{xx}} \sum_{i=1}^n (x_i - \bar{x}) E[y_i | X's]$$

$$= \frac{1}{s_{xx}} \sum_{i=1}^n (x_i - \bar{x}) E[y_i | x_i]$$

strong
exogeneity

A2 $E[y_i | x_i] = \beta_0 + \beta_1 x_i$

↓
Not true for
Task 6

Task 4

The only thing missing in $V(\hat{\beta} | X's)$

$$V(\hat{\beta} | X's) = \begin{bmatrix} \text{Var}(\hat{\beta}_0 | X's) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) & \text{Var}(\hat{\beta}_1 | X's) \end{bmatrix}$$

Recall that Thm 3.3 in Lecture 3 part 3, I computed $\text{Var}(\hat{\beta}_1 | X's)$

$$\begin{aligned} \text{Var}(\hat{\beta}_0 | X's) &= \text{Var}(\bar{Y} - \hat{\beta}_1 \bar{X} | X's) \\ &= \text{Var}(\bar{Y} | X's) + \bar{X}^2 \text{Var}(\hat{\beta}_1 | X's) - 2\bar{X} \text{Cov}(\bar{Y}, \hat{\beta}_1) \end{aligned}$$

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | X's) = \text{Cov}(\bar{Y} - \hat{\beta}_1 \bar{X}, \hat{\beta}_1 | X's)$$

Tools 7

$$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} | X's \sim N_2 \left(\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \underbrace{\frac{\sigma^2}{s_{xx}} \begin{bmatrix} \frac{1}{n} \sum x_i^2 - \bar{x} & \\ & -\bar{x} & 1 \end{bmatrix}}_{\text{Var}(\hat{\beta} | X's)} \right)$$

$$V(\hat{\beta} | X's) = \begin{bmatrix} \text{Var}(\hat{\beta}_0 | X's) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | X's) \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | X's) & \text{Var}(\hat{\beta}_1 | X's) \end{bmatrix}$$

What is $\hat{\beta}_1$ dist. ? $\hat{\beta}_1 \sim N(\beta_1, \text{Var}(\hat{\beta}_1 | X's)) \sim N(\beta_1, \frac{\sigma^2}{s_{xx}})$

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{s_{xx}}}} = \frac{\hat{\beta}_1 - \beta_1}{\frac{\sigma}{\sqrt{s_{xx}}}} \sim N(0, 1)$$